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Modeling and simulation of the material removal process in electrolyte jet machining of mass transfer in convection and electric migration

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Abstract

Electrochemical jet machining (Jet-ECM) is a non-contact machining process which employed jetted column electrolyte as tool to impinge on the workpiece surface. Throughout the developments of this technique, the research on removal mechanism about mass transfer in Jet-ECM is of few interests. This paper presents a multi-physical approach for modeling the material removal process by coupling electrochemical reactions, fluid dynamics and mass transfer as a consequence of diffusion, convection and migration of ions in electrolyte and electrolyte-electrode interface. The electrolyte flow field is calculated for incompressible flow by Navier-Stokes equations. The distribution of ions are found to be a significant factor to current density distribution and geometry change. The consideration of mass transfer is closer to reality.

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Keywords: , Electrolyte jet machining, Multiphysics, Fluid dynamics, mass transfer, Current density distribution

1. Introduction

In the last few decades, electrochemical micromachining (ECM) is widely employed to create micro structures. ECM as a non-contact, no tool wear, and athermal progress, thus free of mechanical and thermal residue stress therefore won't initiate material property change[1]. Material removal rate could be adjusted by controlling the current applied. A wide range of metal materials, regardless hardness and thermal resistance, can be easily machined by ECM. A number of techniques based on ECM are viable to produce micro strucures, such as jet electrochemical machining.

Jet electrochemical machining(jet-ECM) employs electrolyte jet, a stream of electrolyte flows through nozzle and impinges directly on the workpiece, to localized the electrochemical reaction. Jet-ECM has demonstrated the capacity to machine various materials down to the micrometer range in [3,4], even the capability of microstructuring carbide metals [6].

Jet-ECM is a promising method to generate surface textures but suffers from relatively slow speed because of the sequential processing. Costa [2] employed textured tools consisting of arraying holes to fabricate micro dimples by Jet-ECM, presented a more efficient parallel machining process .

There are many studies on simulation of material removal process in Jet-ECM, but most of them are based on potential thus the current **i** was calculated by Ohm's law ($\mathbf{i} = \sigma\phi$), which cannot match the experimental data accurately. It is difficult to simulate the "islands"[5] in actual machining.

In the present study a model was developed with COMSOL Multiphysics to simulate the transportation of ions in convection in electrolyte flow field and the migration in electric field, i.e. the electrodynamics for a more realistic model.

2. Modeling of proposed method with consideration of and materials

2.1. Physical model

The model consists of couplings of electrochemical reactions, ion transportation, fluid dynamics and deformed geometry. Fig. 1 shows the 2D axially symmetric model. The system of equations to be solved consists of the charge conservation8, conservation for incompressible flow and ion transport.

2.1.1. Electrolyte fluid dynamics

The electrolyte flow field is found to be the most influential factor in ECM process, ion transport during the electrochemical process is greatly affected by the electrolyte flow field. It is therefore of utmost importance to have an accurate flow field calculation The electrolyte flow is governed by the lami-

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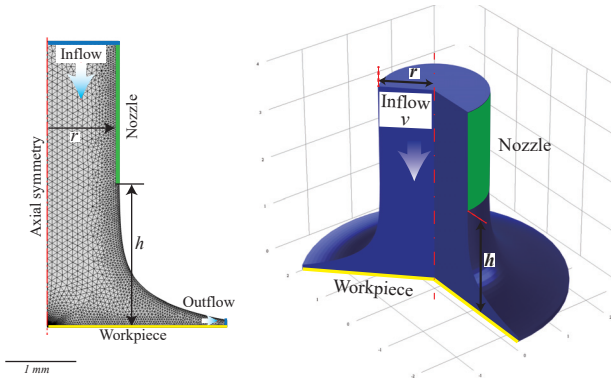

Fig. 1. 2D axially symmetric model.

$$\begin{cases} \rho\frac{\partial \mathbf{u}}{\partial t} + \rho(\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \mu\Delta\mathbf{u} \\ \frac{\partial \rho}{\partial t} + \rho\nabla\cdot\mathbf{u} = 0 \end{cases} \tag{1}$$

Where p is the pressure, ρ the density and $\mathbf{u}$ the velocity of the fluid.

The boundary conditions are:

- Inflow: a constant velocity is imposed:

$$\mathbf{u} = \mathbf{u_0} \tag{2}$$

- Outflow: a pressure of 0 is imposed:

$$p_t = 0 \tag{3}$$

- The electrolyte-electrode boundary interfaces are subjected to a no-slip conditions:

$$\mathbf{u} = 0 \tag{4}$$

2.1.2. Electrodynamics

The potential distribution ϕ is given by the Laplace's eqation:

$$\nabla(\nabla\sigma\phi) = 0 \tag{5}$$

Dissolved species are carried by the electrolyte flow and consumed or produced leading to non-uniform concentration profiles.

$$R_i = \frac{\partial c_i}{\partial t} + \nabla\cdot(-D_i\nabla c_i - \frac{FD_iz_ic_i}{RT}\nabla\phi) + \mathbf{u}\cdot\nabla c_i \tag{6}$$

where R_i the production rate of specie i, F the Faraday constant, D_i the diffusion coefficient, z_i the valency, c_i the concentration of specie i.

The total flux N_i of specie i is determined by Eq. 7:

$$\mathbf{N_i} = -D_i\nabla c_i - \frac{FD_iz_ic_i}{RT}\nabla\phi + \mathbf{u}c_i \tag{7}$$

Assuming the electrolyte conductivity is so large that free charges do not exist:

$$\sum z_ic_i = 0 \tag{8}$$

where z_i is the valency, c_i the concentration of specie i. Current density $\mathbf{i}$ is a result of ion transportation:

$$\mathbf{i} = F\sum z_i\mathbf{N_i} \tag{9}$$

substitution of Eq. 7,8 into Eq. 9 yields:

$$\mathbf{i} = -\frac{F^2}{RT}\nabla\phi\sum D_ic_iz_i^2 - F\sum z_iD_i\nabla c_i \tag{10}$$

where R is molar gas constant, T the temperature of circumstance which is set 293.15K.

The boundary conditions are:

- inflow:

$$c_i = c_0 \tag{11}$$

- outflow:

$$\mathbf{u}\cdot D_i\nabla c_i = 0 \tag{12}$$

- on the cathode(nozzle) the electric potential is ground potential:

$$\phi = \phi_c = 0 \tag{13}$$

water molecules is dissociated into hydroxyl ion and hydrogen gas:

$$2H_2O + 2e^- \rightarrow H_2 + 2OH^- \tag{14}$$

- on the anode(workpiece), oxidation of water is processing, and metal is electrochemical dissolved by applied electric potential:

$$\phi = \phi_a \tag{15}$$

$$Me + 2H_2O \rightarrow Me^{z+} + O^2 + 4H^+ + (z+4)e^- \tag{16}$$

- on the electrolyte-electrode interface, ions are produced or consumed by electrochemical reactions:

$$-\mathbf{N_i}\cdot\mathbf{n} = \sum_m R_i \tag{17}$$

- The geometric deformation implements the material removal process which can be quantified by Eq. 18, the anode removal rate v_a is obtained using Faraday's law.

$$v_a = \eta\cdot\mathbf{i}\cdot\mathbf{n}\frac{M}{Fz_i\rho} \tag{18}$$

where M is the molar mass, η is the current efficiency.

2.2. Simulation conditions

The numerical simulation conditions of 10s are shown in Table 1 and diffusion parameters of ions participated in this model are shown in Table 2.

3. Results and discussion

3.1. Electrolyte fluid dynamics

Fluid dynamics is significant to the Jet-ECM process, since all other important process such as ion transport and fresh elec-

Table 1. Simulation conditions.

Workpiece	Iron plate
Electrolyte	1 mol/L $NaNO_3$ aq
$\mathbf{u_0}$	1 m/s
c_0	1 mol/L
ϕ_a	100V
r	1 mm
h	2 mm

Table 2. Diffusion parameters of ions at 293.15K

	z_i	$D_i(10^{-9}m^2/s)$
Na^+	1	1.33
NO_3^-	-1	1.9
H^+	1	9.31
Me^{z+}	3	1.24
OH^-	-1	5.26

trolyte supply. In the center area of workpiece the flow velocity is slow, whereas the velocity is fast at the rim. The maximum velocity is found at the out let and up to 2.55 m/s; the minimum velocity is found at the center of the workpiece and as low as 0. The electrolyte flow field exerts considerable influence on the machined workpiece shape, more metal removed at where the flow rate is faster.

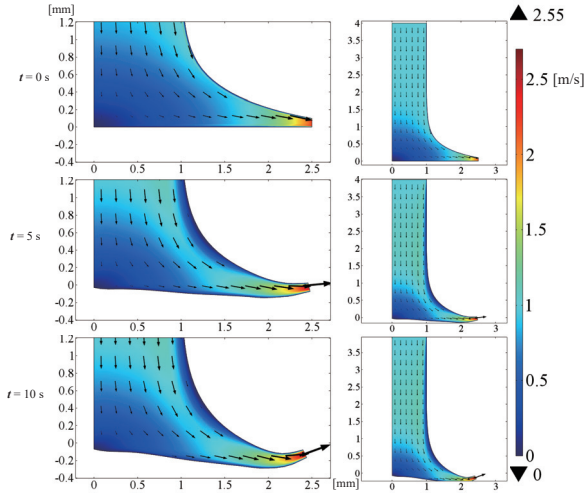

Fig. 2. Simulated velocity **u** of the electrolyte fluid

3.2. Electrodynamics

Fig. 4 shows the ion concentration at workpiece near field. Due to the convection in flow filed shown in Fig. 2, the ion concentration is lower at workpiece surface for lower electrolyte flow rate. The Na^+ ions are positively charge, is repelled by anode (workpiece) and NO_3^- ions are negatively charged, is attracted, this will result a little more NO_3^- ions at the electrolyte -electrode interface where the electric migration is the preponderant power. But as a whole, the convection of ions is the dominant factor to concentration distribution.

The current density **i** as shown in Fig. 5 is lower at the

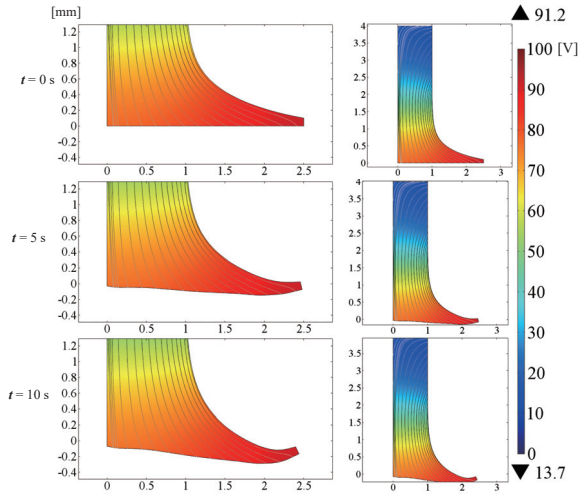

Fig. 3. Electric potential distribution in inter-electrode gap, stream lines represent the electric current density

rim and center of the workpiece and extreme high (up to 107 A/cm^2) right below the nozzle but reduce rapidly before propagate at the workpiece surface. This can be explained by that the current density **i** is determined by the concentration profiles as illustrated in Eq. 10, the nature of **i** deduced by ion transportation, change fast and relatively evenly distributed at workpiece surface.

3.3. Comparison of simulation between the electrodynamics model with the potential model

Fig. 6,9 demonstrate the simulated electric potential and corresponding current density distribution in potential model at different times, the simulation conditions kept the same, the stream lines represent the electric current density, and Fig. 8 shows the comparison between the potential model and electrodynamics model.

The maximum value of current density is up to 219 A/cm^2 in potential model, and mainly distributed in the center of the workpiece, which leads concave in the center. On the contrary, in electrodynamics model, the distribution of current density is relatively uniform but lower at the rim and the center which results in bottom heave i.e. the "island" in actual processing[5].

4. conclusion

In this study a multiphysical model for the simulation of the jet-ECM process was created by coupling electrochemical reactions, fluid dynamics and mass transfer as a consequence of diffusion, convection and migration of ions in electrolyte and electrolyte-electrode interface was performed. The electrolyte flow field is calculated for incompressible flow by Navier-Stokes equations. The distribution of ions are found to be a significant factor to current density distribution and geometry change. The concentration distributions of the corresponding electric current density distribution were obtained. The island[5] in actual machining can be simulated in present work. The comparison of the model in present paper with the poten-

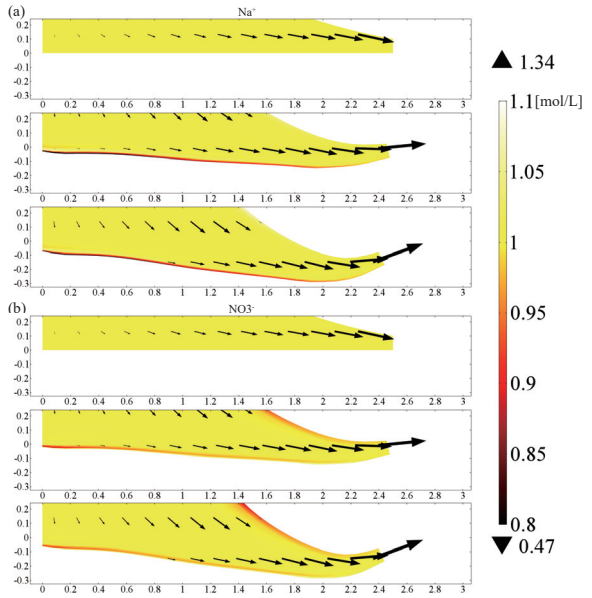

Fig. 4. Concentration distribution at workpiece near field. (a) Na^+ ion concentration; (b) NO_3^- ion concentration

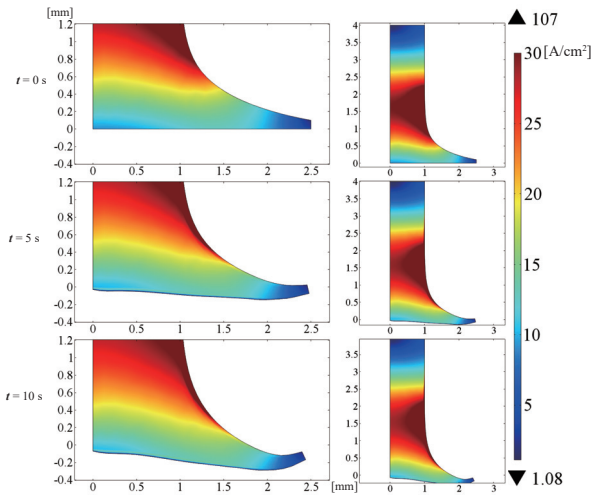

Fig. 5. Electric current density

tial model demonstrated that the consideration of ion transport equations will be closer to reality.

Acknowledgments

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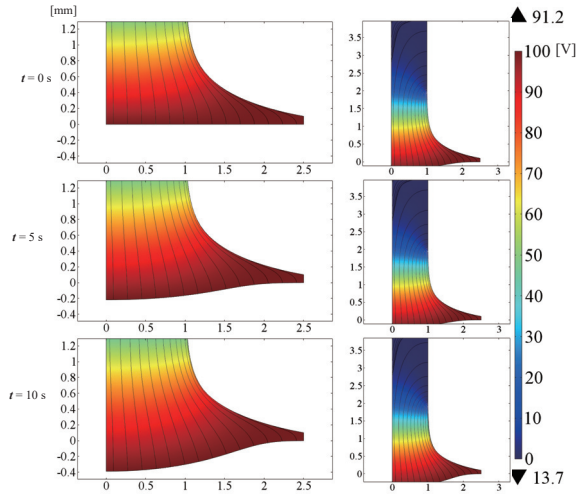

Fig. 6. Electric potential distribution simulated in potential model, stream lines represent the electric current density

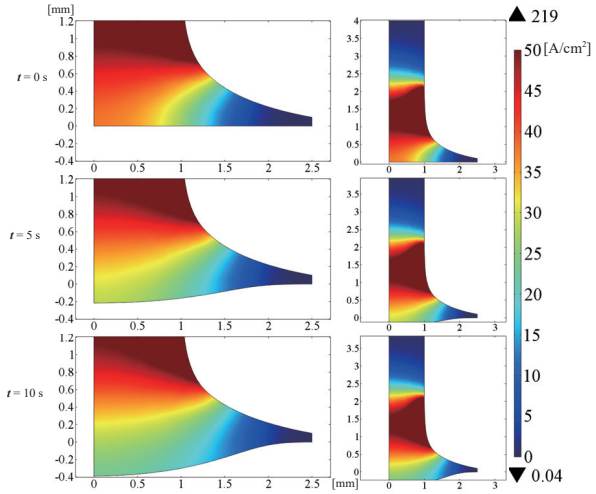

Fig. 7. Electric current density simulated in potential model

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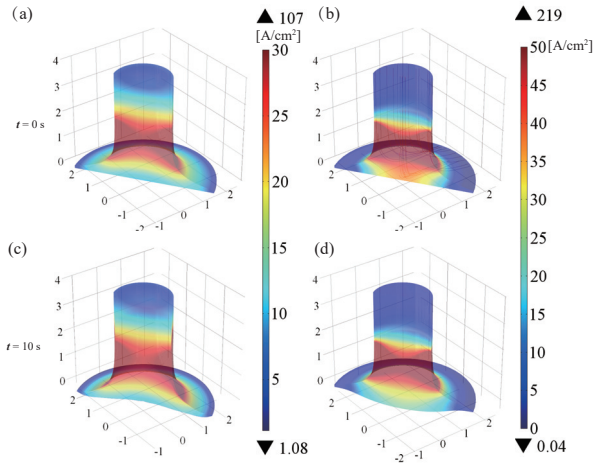

Fig. 8. Comparison of simulation between the electrodynamics model with the potential model.(a) Electric current density distribution of electrodynamics model at t=0 s; (b) Electric current density distribution of potential model at t=0 s; (c) Electric current density distribution of potential model at t=10 s; (b) Electric current density distribution of potential model at t=10 s

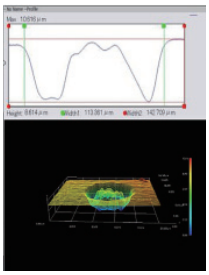

Fig. 9. “Island” in electrochemical machining[5]